Analysis of Emotions and Stress Level with DeepFace

Sapienza Università di Roma

A.A 2024-2025

Biometric Systems Project

Facoltà di Ingegneria dell'Informazione, Informatica e Statistica
Dipartimento di Informatica
Corso di laurea in Computer Science

Valerio Mesiti - 1936543

Gabriele Moretti - 1958932

Summary

Abstract 3
Chapter 1: Introduction 4
1.1 Context and Motivation 4
1.2 Purpose and Objectives 4
Chapter 2: Review and Background 5
2.1 Emotion Recognition 5
2.2 DeepFace 6
2.3 Datasets Used 6
2.3.1 AffectNet 7
2.3.2 RAF-DB 8
Chapter 3: Methodology and Implementation 9
3.1 Data Preparation 9
3.1.1 Evaluation Metrics 9
3.2 Data Analysis Script 11
3.2.1 Data Saving 11
3.2.2 Emotion Detection 12
3.2.3 Stress Level Calculation 13
3.2.4 “Not Detected” 14
3.3 Classification Script and Data Report 14
Chapter 4: Results 16
4.1 Val_Set Analysis 16
4.1.1 First Analysis 16
4.1.2 Enforced Analysis Not Detected 20
4.1.3 Total Enforced Analysis 23
4.2 RAF_DB Analysis 23
4.2.1 First Analysis 24
4.2.2 Enforced Analysis Not Detected 28
4.2.3 Total Enforced Analysis 31
4.3 Comparison between the two Datasets 31
4.4 Limitations and Future Developments 32
Chapter 5: Conclusions 33
5.1 Summary of Results 33
5.2 Research Impact 33
5.3 Challenges Overcome 33
5.4 Closure 33
Bibliography 34

Abstract

This report presents an **analysis of human face** images with the aim of **assessing** the stress level associated with the emotions expressed at the time of capture.

Human face images are taken from the pre-existing **AffectNet** and **RAF-DB** datasets and are analyzed using the **DeepFace** framework for emotion recognition. An **evaluation of the effectiveness** of DeepFace in correctly distinguishing and classifying different facial expressions was also conducted.

Based on the detected emotions, an indicator was developed that classifies the **stress level** into **three categories**: low, medium and high.

The results show that, although DeepFace is **effective** in recognizing some emotions, it has **limitations** in distinguishing similar emotional states, with implications on the **accuracy** of stress classification.

The analysis also highlights the potential of this methodology in monitoring stress in application contexts, such as **recognizing states of fatigue** or **risky behaviors**.

Finally, possible strategies to improve the accuracy of the model are discussed, including **dataset optimization** and **integration with more advanced** facial expression analysis **models**.

Chapter 1: Introduction

1.1 Context and Motivation

Emotions play a fundamental role in **human communication** and **social perception**. Understanding and accurately classifying emotions can have relevant applications in areas such as **healthcare**, **monitoring work stress**, and **psychological research**. However, the complexity and subjectivity of emotions pose a **challenge** to machine learning models, making universal classification particularly difficult, especially in unstructured contexts.

An innovative application of emotion recognition concerns the **monitoring of behaviors in gambling contexts**. The idea behind this project comes from the intent to **evaluate a system** capable of analyzing emotions in order to examine and **classify the level of stress**. The goal is to identify signs of compulsive stress, such as **intense anger or fear**, in order to intervene promptly and **prevent potential problems** related to addiction.

1.2 Purpose and Objectives

The project aims to **evaluate the accuracy** of the **DeepFace** framework in recognizing emotions on images of human faces and define a criterion to **quantify the level of stress** manifested by an individual based on his or her **facial expression**.

The **objectives** of the project, therefore, can be defined as follows:

- **Verify the behavior of the DeepFace framework** by examining, for each emotion defined by the dataset, the number of images in which the emotion is **correctly recognized**, the number of images in which the **recognition is incorrect**, the number of images in which the expression is **not detected** and the related **reasons**;
- Use **two datasets** for the study of facial expression recognition, characterized by **different image sizes** and **complexity** (in terms of variety of poses, lighting conditions, quality and contexts), in order to evaluate the performance of the DeepFace framework on images with **heterogeneous characteristics**;
- **Develop a methodology** to quantify perceived stress based on the main detected emotions, through a process of **reviewing and labeling** the datasets used;
- **Analyze the effectiveness** of the methodology developed for the determination of the stress level in the context under consideration;
- **Identify the limitations of the framework**, propose future developments to improve accuracy and robustness.

Chapter 2: Review and Background

2.1 Emotion Recognition

Emotion recognition is a multidisciplinary field that combines the principles of **Computer Vision**, **Artificial Intelligence** and **Natural Language Processing** to identify and analyze human emotional states. This process is based on the extraction and interpretation of signals from different sources, including: **facial expressions**, **tone of voice**, **body movements** and **physiological parameters** such as heart rate and sweating. The main methodologies used for emotion recognition can be divided into three categories:

1. **Image and video analysis**: facial expressions are identified through **Deep Learning** algorithms, in particular **Convolutional Neural Networks** (CNN), which analyze the muscular movements of the face (e.g. smiles, frowning);
2. **Speech analysis**: the tone, timbre and intonation of the voice provide crucial information about emotions. **Natural Language Processing** (NLP) and **Machine Learning models** can detect emotions such as anger, happiness or sadness through audio analysis;
3. **Analysis of physiological signals**: biometric sensors monitor parameters such as heart rate, sweating and brain activity to determine emotional states. This technique is particularly useful in the medical and psychological fields.

Emotion recognition technologies find application in numerous sectors:

- **Security and surveillance**: analysis of facial expressions to identify signs of stress or suspicious behavior;
- **Human-machine interaction**: improving the user experience in virtual assistants and social robots;
- **Mental health and well-being**: monitoring the emotional states of patients with mood disorders or autism spectrum disorders;
- **Marketing and customer experience**: analysis of consumers' emotional reactions to optimize advertising campaigns and commercial strategies.

Despite technological progress, automatic emotion recognition still has some critical issues:

- **Ambiguity in facial expressions**: the same gesture can be interpreted in different ways depending on the context and the individual's culture;
- **Model generalization**: Model accuracy can vary based on factors such as ethnicity, age, and lighting conditions;
- **Privacy and ethical issues**: Emotion tracking raises concerns about data protection and ethical use of these technologies.

2.2 DeepFace

DeepFace is a pre-trained model developed by **Meta** for facial recognition, designed to **identify and verify** human faces with high accuracy. Open-source implementations of the DeepFace methodology are available, such as the DeepFace library for **Python**, which is used for emotion analysis, facial recognition, and other image processing-related applications.

The application of DeepFace in emotional analysis is an extension of its facial recognition capabilities, leveraging **deep learning** techniques to **detect and classify** human emotions based on facial expressions.

The framework performs a **process of face detection and alignment** in an image, ensuring that the **position, orientation, and size** are uniform, thus optimizing the accuracy of the analysis. DeepFace uses **convolutional neural networks** (CNNs) to extract significant facial features, such as **eyes, mouth, and eyebrows**, which play a critical role in the **expression of emotions**. Starting from these features, the model is able to classify the emotions of the analyzed face into different categories, including: **happiness, sadness, fear, anger, surprise, disgust and neutral**.

In addition, DeepFace is designed to **ensure high robustness** in face analysis, demonstrating the ability to operate effectively in **variable environmental conditions**, such as changes in **lighting, different angles of the face** and **non-uniform image quality**.

In unstructured environments or with low-quality images, DeepFace performance may **decrease**.

2.3 Datasets Used

For the evaluation of the emotion recognition model, two datasets widely recognized in the scientific community were used: **AffectNet** and **RAF-DB**. These datasets contain images of faces labeled with different emotional categories, providing a wide variety of expressions and acquisition conditions.

The combined use of these two datasets allows to obtain a more generalizable model, capable of recognizing emotions in real contexts. This approach helps mitigate issues related to **imbalances or biases in the data** and improves the ability to effectively distinguish between different emotional classes.

2.3.1 AffectNet

AffectNet is one of the largest and most comprehensive datasets available for analyzing emotions and facial expressions. It includes over a **million facial images** automatically collected through search engines, using queries based on emotion-related keywords. A subset of the dataset, consisting of approximately 450,000 images, has been manually annotated to ensure greater accuracy.

The images are classified into **eight main emotional categories**: neutral, happy, sad, surprised, fear, disgust, anger and an additional category called "**other**", which includes facial expressions that cannot be traced back to the above-mentioned emotions.

The images are provided in RGB format (in color) but are not uniformly sized.

Additional information is available for each image:

- **Facial landmarks**: up to **68 key points** (e.g. eyes, nose and mouth) to identify and track the salient features of the face;
- **Valence**: A numerical value representing how positive or negative an emotion is, on a scale from -1 (very negative) to +1 (very positive);
- **Arousal**: A value that measures the intensity or arousal level of the emotion, on a scale from calm (-1) to excited (+1);
- **Bounding box**: Coordinates to locate the face within the image.

The images come from natural contexts and have a wide range of variations in terms of acquisition conditions: **lighting** (day, night, artificial light); **poses** (frontal, profile, tilt); **age** (from infants to elderly); **gender** and **ethnicity**. This variety allows the dataset to better reflect real-world conditions.

AffectNet is divided into two main sets:

- **Training set**: Consisting of approximately **450,000 manually annotated images**, categorized by emotion;
- **Test set**: Consisting of approximately **500,000 automatically annotated images**.

The distribution of emotions in the dataset is not uniform: **happiness** is the most represented emotion, followed by **neutral**. Emotions such as **fear** and **disgust** are significantly less present. This disparity can negatively affect the performance of the models in recognizing the less represented categories.

Since the images are collected from the Internet, the dataset can be affected by demographic or cultural imbalances, limiting generalizability in global contexts.

Furthermore, the classification of emotions is influenced by the subjectivity of the annotators, which could introduce interpretation errors.

2.3.2 RAF-DB

RAF-DB (Real-world Affective Faces Database) is a widely used dataset in the field of facial emotion recognition, containing about **30,000 facial images** collected from the Internet. The images are labeled by human annotators based on the emotions perceived in the faces, classifying the emotions into seven main categories: **neutral, happy, sad, surprised, fearful, disgusted, and angry.**

The dataset offers a wide demographic representation, with images covering different **ages** (from infants to elderly), **genders**, **ethnicities** (including individuals from different regions of the world), and **environmental conditions** (such as variations in lighting, backgrounds, image quality, and face angles). Such diversity makes it suitable for analyses on global populations.

The images come from natural contexts, capturing spontaneous and authentic expressions, and are provided in **RGB format**, with uniform dimensions.

RAF-DB also provides **facial landmarks**, i.e. key points of the face (eyes, nose, mouth, etc.), which can be used to align faces and track their salient features, thus improving the performance of deep learning models.

The dataset is split into **training** and **test sets**, with a predefined proportion of images assigned to each set to facilitate validation and training.

However, as in the case of the AffectNet dataset, emotions are not uniformly distributed: emotions such as “happiness” and “neutral” are overrepresented, while emotions such as “fear” and “disgust” are less frequent, creating an imbalance in the data and impacting the performance of the models, especially for the less represented emotions.

Furthermore, since emotions are labeled based on human perception, there may be discrepancies due to cultural or interpretative differences between annotators.

Chapter 3: Methodology and Implementation

3.1 Data Preparation

The **AffectNet** test set, called "**val_set**", and the entire **RAF-DB** dataset were taken into consideration. The images in the datasets were organized and divided into seven main emotional categories: **Happy, Sad, Angry, Fear, Surprise, Disgust, Neutral.**

DeepFace was used to analyze **each image**, returning the predominant emotion and the relative probabilities associated with each emotional category.

3.1.1 Evaluation Metrics

Although emotion recognition falls within the macro-category of biometric systems, the metrics traditionally used to evaluate the accuracy of classic biometric systems, such as the **False Acceptance Rate (FAR)** and the **False Rejection Rate (FRR)**, are not applicable in this context. These parameters are specifically used in authentication systems, where it is essential to minimize errors in accepting or rejecting an identity.

In the case of emotion recognition, however, the problem is configured as a typical multi-class classification task, in which the model must distinguish between different emotional categories rather than verify the identity of an individual. Consequently, standard Machine Learning metrics are more appropriate for evaluating the system performance.

For the performance evaluation of the emotion recognition model, standard metrics of Machine Learning systems were adopted, including:

- **Accuracy**: measures the percentage of correct classifications compared to the total samples analyzed. It is calculated using the following formula:

$$Accuracy = \frac{True\ Positives + True\ Negatives}{Total\ Samples}$$

  where:

  - **True Positives (TP)**: number of samples correctly classified as belonging to a class;
  - **True Negatives (TN)**: number of samples correctly classified as not belonging to a class;
  - **Total Samples**: total number of elements in the dataset.

- **Precision**: measures the proportion of items assigned to a given class that are actually correct. It is calculated using the following formula:

$$Precision = \frac{TP}{TP + False\ Positive}$$

  where:

  - **True Positives (TP)**: number of samples correctly classified as belonging to a specific class;
  - **False Positives (FP)**: number of samples incorrectly classified as belonging to the class under examination.
- **Recall**: measures the model's ability to correctly recognize all samples belonging to a given class. It is calculated using the following formula:

$$Recall = \frac{TP}{TP + False\ Negative}$$

  where:

  - **True Positives (TP)**: number of samples belonging to a class that were correctly recognized;
  - **False Negatives (FN)**: number of samples belonging to the class that the model did not recognize correctly.
- **F1-Score**: represents the harmonic mean between Precision and Recall, offering a balanced indicator that takes into account both the model's ability to correctly identify the samples of a class (Recall) and its precision in avoiding incorrect classifications (Precision). It is calculated with the following formula:

$$F1\ Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

- **Confusion matrix:** a table that allows you to evaluate the performance of a classification model by showing how the model's predictions compare to the actual labels. Each row represents instances of the actual class, while each column indicates instances of the class predicted by the model.

These metrics allow us to evaluate the effectiveness of the model in correctly differentiating emotions and minimizing classification errors.

The analysis of the Confusion Matrix is fundamental to understand the errors made by the model and to improve its ability to discriminate between different classes.

3.2 Data Analysis Script

The emotion analysis was performed using the DeepFace library, which allows to detect the **dominant emotion** in an image and to provide the **percentage distribution** of the recognized emotions.

The code below illustrates the **image analysis** process, including **emotion detection**, **misclassification management** by checking the correspondence between the detected emotion and the **predicted label**, as well as the **calculation of the stress level** based on the distribution of emotions. Furthermore, the analysis results are recorded in a **CSV file** for further processing.

```
#funzione di rilevamento emozioni
emotions = DeepFace.analyze(img, ['emotion'])[0]

# Verifica se l'emozione rilevata corrisponde all'etichetta prevista
# In caso di errore, l'immagine viene copiata nella cartella degli errori
if(emotions['dominant_emotion'] != emotion_folder):
      shutil.copy(image_path, os.path.join(emotion_error_emotion,
file_name))

#calcolo del livello di stress
stress_level = calculate_stress(emotions['emotion'])

#scrittura sul file CSV principale
writer1.writerow({
      "file_name": file_name, #Nome del file immagine analizzato
      "dominant_emotion": emotions['dominant_emotion'], #Emozione dominante
rilevata
      "emotions": {key: round(value, 3) for key, value in
emotions['emotion'].items()}, # Distribuzione delle emozioni con 3 decimali
      "stress_level": stress_level # Livello di stress calcolato })
```

3.2.1 Data Saving

For each processed image, the model generated **CSV files** containing detailed information such as the **photo ID**, the **predominant emotion**, the **distribution of detected emotions** with their **percentages** and a **calculated stress level**. This data structure made it easier to analyze and evaluate the model's performance.

The following screenshot shows an extract of the content of the CSV file generated by the analysis of images belonging to the category labeled as “Happy”.

```
file_name,dominant_emotion,emotions,stress_level
101.jpg,happy,"{'angry': 0.0, 'disgust': 0.0, 'fear': 0.357, 'happy': 51.125, 'sad': 0.096, 'surprise': 0.128, 'neutral': 48.295}",Low Stress
1012.jpg,happy,"{'angry': 0.001, 'disgust': 0.0, 'fear': 0.044, 'happy': 82.966, 'sad': 0.179, 'surprise': 0.0, 'neutral': 16.81}",Low Stress
1043.jpg,happy,"{'angry': 0.0, 'disgust': 0.0, 'fear': 0.0, 'happy': 99.827, 'sad': 0.005, 'surprise': 0.0, 'neutral': 0.167}",Low Stress
1052.jpg,happy,"{'angry': 0.0, 'disgust': 0.0, 'fear': 0.0, 'happy': 99.901, 'sad': 0.0, 'surprise': 0.0, 'neutral': 0.099}",Low Stress
1085.jpg,neutral,"{'angry': 0.007, 'disgust': 0.0, 'fear': 0.0, 'happy': 0.03, 'sad': 0.007, 'surprise': 0.0, 'neutral': 99.956}",Low Stress
1086.jpg,happy,"{'angry': 0.0, 'disgust': 0.0, 'fear': 0.0, 'happy': 99.855, 'sad': 0.0, 'surprise': 0.0, 'neutral': 0.145}",Low Stress
1087.jpg,happy,"{'angry': 0.0, 'disgust': 0.0, 'fear': 0.0, 'happy': 98.994, 'sad': 0.0, 'surprise': 0.0, 'neutral': 1.006}",Low Stress
1089.jpg,happy,"{'angry': 0.0, 'disgust': 0.0, 'fear': 0.011, 'happy': 98.279, 'sad': 0.006, 'surprise': 0.008, 'neutral': 1.695}",Low Stress
1096.jpg,happy,"{'angry': 0.748, 'disgust': 0.0, 'fear': 21.397, 'happy': 67.139, 'sad': 2.443, 'surprise': 0.007, 'neutral': 8.265}",Low Stress
```

3.2.2 Emotion Detection

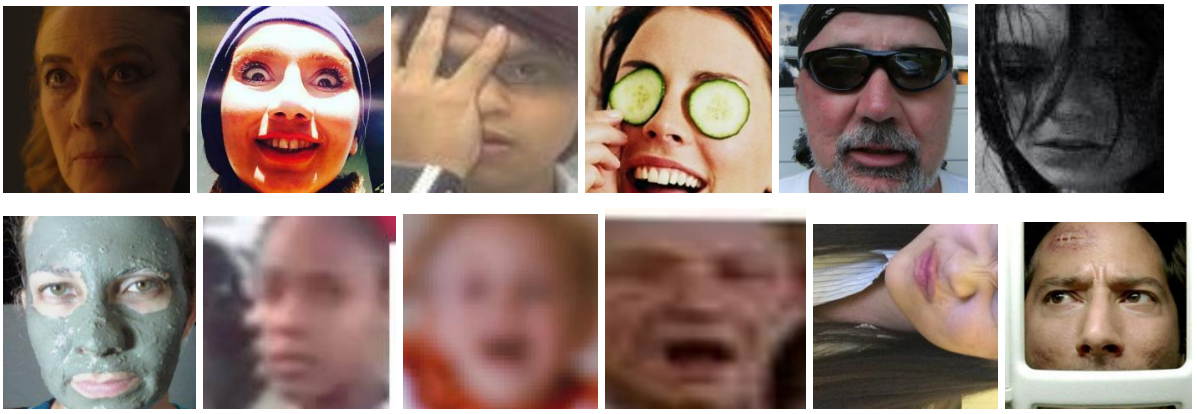
The DeepFace.analyze() method is used to **examine the image** and return a dictionary containing the **predominant emotion**, along with the probabilities associated with **each detected emotion.**

During the facial expression analysis, the model failed to detect some faces, returning the error "Face Not Detected". To address this issue, the **enforce detection** option was enabled, which “forces” the model to detect a face even in difficult conditions.

The main causes of the DeepFace model's failure to detect faces can be attributed to several factors:

- **Lighting issues**: insufficient or excessive lighting conditions can reduce the contrast of facial features, compromising the model's ability to correctly identify them;
- **Partial or total occlusions**: the presence of objects such as glasses, hats, masks or hands in front of the face can prevent face recognition;
- **Image quality**: Blurry, pixelated, or low-resolution images can compromise the extraction of facial features needed for analysis;
- **Partial visibility of the face**: if only part of the face is exposed, the model may not have enough information to accurately identify the full physiognomy.

Below are examples of images, selected from the two datasets used, where the failure to detect faces is attributable to each of the factors previously described.



If the detected emotion does not match the label assigned to the image source folder, it is considered as a classification error and copied to a dedicated folder. This procedure allows for a more in-depth analysis of the types of errors made by the model.

3.2.3 Stress Level Calculation

The idea of this project is to develop, in future advances, a **system capable of detecting the stress level**. Currently, a function has been implemented that calculates the percentages associated with the seven main emotions detected by the model in the face image and, based on predefined parameters in the code, **determines the stress level** by classifying it as low, medium or high. In perspective, the system could be integrated with a **biometric sensor for detecting the heartbeat** or with a **thermal camera** for analyzing the temperature of the different areas of the human face, in order to improve the accuracy of the results.

The function calculate_stress() processes the distribution of the detected emotions to **estimate** the stress level associated with the image. In particular, emotions such as anger and fear tend to indicate a **higher level of stress**, while happiness and neutrality are generally associated with a **reduced state of stress**.

The level of stress is determined based on specific predefined thresholds:

- If the predominant emotion is **happiness** (over 30%) or **neutrality** (over 50%), the level of stress is classified as **low** (Low Stress), since these emotions are generally associated with states of tranquility or well-being;
- If the sum of the probabilities of **fear**, **sadness** and **surprise** exceeds 90%, the level of stress is classified as **medium** (Medium Stress), denoting an emotional state characterized by insecurity or uncertainty;
- If the sum of the probabilities of **anger** and **disgust** exceeds 40%, the level of stress is considered **high** (High Stress), since these emotions are typically associated with situations of strong tension or discomfort;
- If none of the previous conditions are met, the level of stress is labeled as **undefined** (Undefined Stress).

```
def calculate_stress(emotion_probs):
    # Estrapola le percentuali
    happy = emotion_probs.get("happy", 0)
    sad = emotion_probs.get("sad", 0)
    neutral = emotion_probs.get("neutral", 0)
    anger = emotion_probs.get("angry", 0)
    fear = emotion_probs.get("fear", 0)
    disgust = emotion_probs.get("disgust", 0)
    surprise = emotion_probs.get("surprise", 0)

    # Calcola il livello di stress
    if happy > 30 or neutral > 50:
        return "Low Stress"
    elif fear + sad + surprise > 90:
        return "Medium Stress"
    elif anger + disgust > 40:
        return "High Stress"
    else:
        return "Undefined Stress"
```

3.2.4 “Not Detected”

In cases where the face is not detected correctly, the DeepFace model still performs emotion analysis, returning a result based on the available information. Images where the detected dominant emotion does not match the label of the source folder are copied to a specific folder dedicated to classification errors for images analyzed with forced detection.

The analysis results, including the file name, the dominant emotion, the distribution of emotions with values rounded to three decimal places, and the calculated stress level, are written to a secondary CSV file.

All images where the face is not detected correctly are moved to the "**not_detected**" folder, inside a subfolder corresponding to the assigned emotion label.

```
#funzione di rilevamento emozioni enforced
emotions = DeepFace.analyze(img, ['emotion'], enforce_detection = False)[0]

#se l’emozione non corrisponde con l’etichetta la si copia nella cartella
errori
if(emotions['dominant_emotion'] != emotion_folder):
      shutil.copy(image_path,
os.path.join(not_detected_emotion_error_emotion, file_name))

#calcolo del livello di stress
stress_level = calculate_stress(emotions['emotion'])

#scrittura sul file CSV secondario
writer2.writerow({
      "file_name": file_name,
      "dominant_emotion": emotions['dominant_emotion'],
      "emotions": {key: round(value, 3) for key, value in
emotions['emotion'].items()},
      "stress_level": stress_level
})

#spostamento dell’immagine nella cartella “not_detected”
print("Face Not Detected")
shutil.move(image_path, os.path.join(not_detected_folder, emotion_folder,
file_name))
```

3.3 Classification Script and Data Report

To evaluate the accuracy of the emotion recognition model, the main classification metrics previously described were adopted, including: **accuracy**, **precision**, **recall**, **F1-score** and the **confusion matrix**.

The following code describes the comparison process between the real labels (ground truth) and those predicted by the model, using the data contained in the CSV files generated by the image analysis.

```
# Determina l'etichetta vera dalla struttura del nome del file
true_emotion = csv_file.split("_")[-1].split(".")[0]
# Es. da "stress_results_happy.csv" estraiamo "happy"

# Legge il file CSV
with open(os.path.join(csv_folder, csv_file), "r") as csvfile:
      reader = csv.DictReader(csvfile)

      for row in reader:
           file_name = row["file_name"]
           predicted_emotion = row["dominant_emotion"]

           true_labels.append(true_emotion)
           predicted_labels.append(predicted_emotion)
```

The real label is determined by extracting the reference emotional category from the name of the CSV file, which contains the information associated with the correct emotion. Subsequently, the file is read and, for each analyzed image, the real label and the label predicted by the model are stored in two separate lists, respectively for the real and predicted labels, in order to allow subsequent evaluations of the model performance.

The model evaluation metrics are calculated using the functions of the sklearn.metrics library.

```
from sklearn.metrics import confusion_matrix, classification_report,
accuracy_score, precision_score, recall_score, f1_score

# Calcolo delle metriche
accuracy = accuracy_score(true_labels, predicted_labels)
precision = precision_score(true_labels, predicted_labels,
average="weighted")
recall = recall_score(true_labels, predicted_labels,
average="weighted")
f1 = f1_score(true_labels, predicted_labels, average="weighted")

# Calcolo delle metriche
conf_matrix = confusion_matrix(true_labels, predicted_labels,
labels=np.unique(true_labels))
```

Chapter 4: Results

In this section, the **results** obtained on the validation sets are analyzed in order to **evaluate the performance** of the emotion classification model. The analysis focuses on the **confusion matrix** and the **derived metrics**, such as **accuracy**, **precision**, **recall** and **F1-score**.

4.1 Val_Set Analysis

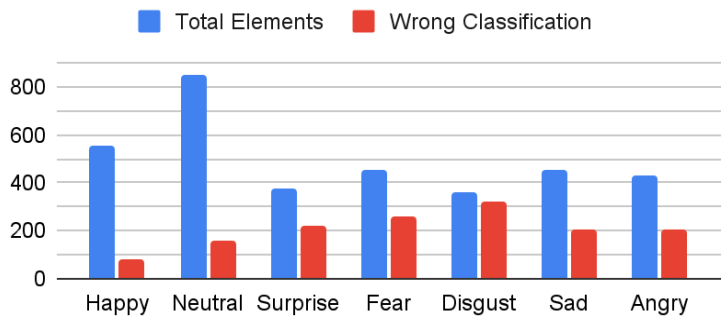
The val_set folder of the AffectNet dataset contains a set of **3481 images** depicting human faces, each associated with a label indicating the predominant emotion. The distribution of the images within the validation set is as follows:

- Happy: 552 images;
- Neutral: 850 images;
- Surprise: 374 images;
- Fear: 456 images;
- Disgust: 361 images;
- Sad: 456 images;
- Angry: 432 images.

This dataset is used to evaluate the performance of the emotion recognition model, **comparing the predicted labels with those provided in the validation set.**

4.1.1 First Analysis

The graph below shows the **total number** of items for each emotion in the validation set (indicated in blue) and the number of **incorrect classifications** (indicated in red).



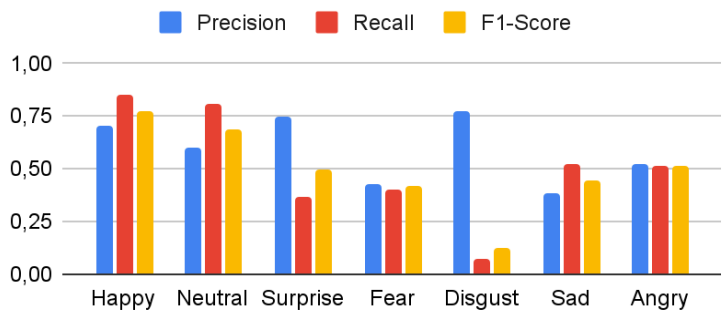
It is observed that the Neutral emotion is the most represented in the dataset, followed by Happy.

The analysis of **classification errors** shows that the **number of incorrect predictions varies significantly depending on the category**. In particular, some emotions, such as Disgust and Surprise, show a particularly **high** error rate compared to the total number of items belonging to these categories, **93.39%** and **62.75%**, respectively. This suggests that the model may have **difficulty** in correctly recognizing these emotions, probably due to **similarities with other categories** or a **lower representativeness** of the distinctive features in the training data.

Other emotions, such as Fear, Sad and Angry, also show a **significant number of errors**, although with **lower classification error rates** than Disgust and Surprise. For example, Fear shows a **significant** number of errors, indicating that the model may confuse it with other emotions such as Surprise or Sad, perhaps due to **similar facial expressions** or **similarities in emotional meaning.** Similarly, Sad and Angry show a **non-negligible error rate**, which may arise from common facial features with other negative emotions, making them more difficult to **accurately recognize**.

In contrast, emotions such as Happy and Neutral show a **lower number of errors** in proportion to the total number of items, suggesting a **higher reliability** of the model in recognizing these categories. This result could be attributed to the **greater distinctiveness of the expressions associated** with these emotions or to a **better representation of these classes** in the training data.

The following graph shows a comparison of the **Precision, Recall and F1-Score** metrics for each emotion, providing an overview of the model's performance in classifying the different categories.



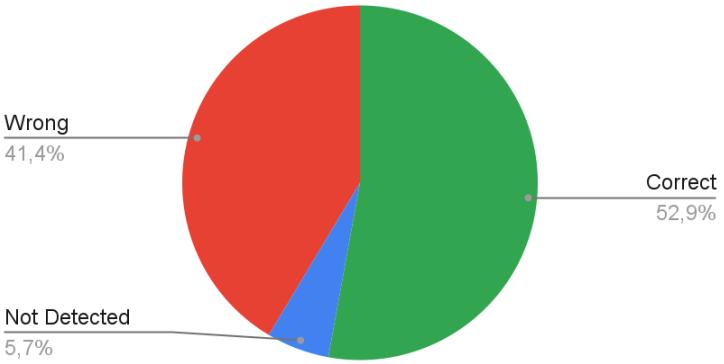
It is observed that the model obtains **better results for the Happy and Neutral emotions**, characterized by relatively high values of Precision and Recall (Happy: **Precision** ≈ 0.7, **Recall** ≈ 0.85; Neutral: **Precision** ≈ 0.5, **Recall** ≈ 0.81). This suggests that the model is able to identify these emotions with a good level of reliability, managing to maintain a balance between the ability to correctly recognize positive instances and the precision of the predictions made.

On the contrary, the model shows **significant difficulties in recognizing some emotions**, in particular **Disgust** and **Fear**. In these cases, the values of Recall and F1-Score are particularly low (Disgust: **Recall** ≈ 0.07, **F1-Score** ≈ 0.12; Fear: **Recall** ≈ 0.4, **F1-Score** ≈ 0.42), highlighting a poor ability of the model to correctly identify these emotions. The low Recall for Disgust suggests that many instances of this emotion are misclassified into other categories, significantly reducing the model's ability to detect it.

For **Surprise**, **Sad**, and **Angry**, the metrics are in **between**, indicating a moderate performance in classifying them. However, we note that Surprise has a lower Recall than Precision, which may indicate that the model tends to correctly identify some instances of this emotion but fails to capture the entire distribution of instances. In contrast, for Sad and Angry, Precision and Recall are more balanced, with an F1-score that reflects a moderate recognition ability.

In general, a lower F1-score indicates a suboptimal balance between Precision and Recall for some categories.

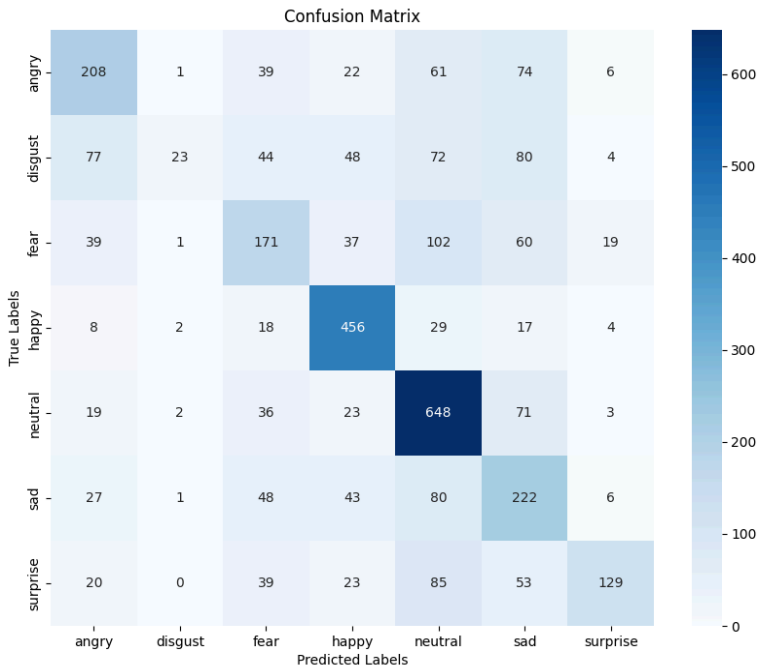
The pie chart below illustrates the overall distribution of classifications performed by the model, distinguishing between **correct classifications** (in green), **incorrect classifications** (in red), and **images where the face was not detected** (in blue).



The data analysis shows that the model **correctly classified** about **52.9%** of the items, while **41.4%** were **incorrectly classified**. Furthermore, **5.7% of the images were not detected**, highlighting potential difficulties of the model in recognizing certain faces or less favorable image acquisition conditions.

The **high percentage of incorrect classifications** suggests that the model may **confuse some emotions with each other**, probably due to **similarities in facial** expressions or an **unbalanced distribution of the training data**. Furthermore, the share of undetected items indicates the need to **improve the pre-processing phase** or the **accuracy of face detection**.

The **confusion matrix** reported below illustrates the distribution of the **model predictions with respect to the real labels**, highlighting the level of accuracy and the main sources of error in emotion classification.



For each emotion, we observe that:

- **Angry**: the model **correctly classified 208 instances**, but showed some **confusion with the emotions sad, neutral and fear**, suggesting that certain expressions of anger could be misinterpreted as other negative emotions;
- **Disgust**: **only a few instances were correctly recognized**, with a significant **dispersion of errors between the sad, angry and neutral** categories, highlighting a general difficulty in distinguishing this emotion, probably due to a lower representativeness in the training data or similarities with other categories;
- **Fear**: the model **assigned 171 instances to the correct category**, but a **significant amount of examples were incorrectly classified as neutral and sad**, suggesting a possible difficulty of the model in distinguishing expressions of fear from more neutral or melancholic emotional states;
- **Happy**: this is one of the best recognized emotions, with **456 correctly classified instances**. The model, however, presents some **confusion with neutral, fear and sad**, indicating that some happy expressions may share characteristics with other emotions;
- **Neutral**: it is the most identified emotion with **648 correct classifications**, indicating a good ability of the model to recognize it. However, **significant errors are observed with sad, fear and happy**, which suggests that some neutral expressions may be interpreted as slightly positive or negative;
- **Sad**: the model **correctly identified 222 instances**, but showed a non-negligible **confusion with neutral, fear and happy**, highlighting that some sad expressions may visually overlap with those of other emotions;
- **Surprise**: the model **correctly classified 129 instances**, but showed a certain tendency to **confuse this emotion with neutral and sad**, suggesting a potential similarity in expressive characteristics.

Overall, the model showed a **good ability to recognize emotions** such as happy and neutral, while it had **difficulties** with disgust and fear, highlighting room for improvement in distinguishing similar or less represented emotions in the dataset.

The results of this analysis suggest possible improvement strategies in training and optimizing the model, such as **balancing the dataset, refining the extracted features** for the classes most subject to classification errors, or **improving the architecture** in order to reduce confusions between visually similar emotions.

4.1.2 Enforced Analysis Not Detected

The total number of images in which the face was not detected amounts to **199**. For a part of these, where possible, the cause of the failure to detect was identified, attributing it to the problems reported in the following table:

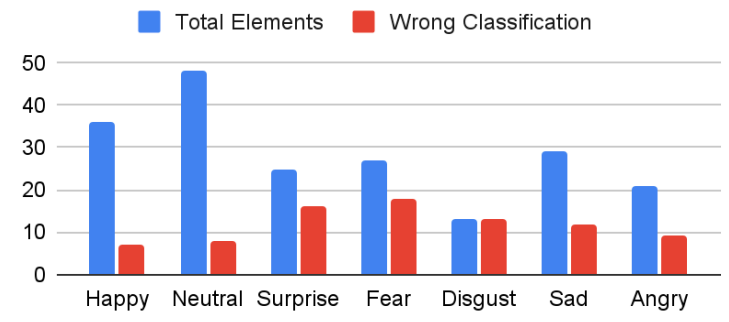
Reason for failure to detect	**Number of Images**
Lighting problems	20
Partial or total occlusions	45
Poor image quality	2
Incomplete exposure of the face	29

The analysis shows that the **most frequent reason for missed detection** is represented by **partial or total occlusions, which account for approximately 22.6% of the total cases**. **Incomplete facial exposure** (approximately 14.6%) and **lighting problems** (approximately 10.1%) follow. The small number of images affected by **insufficient quality** suggests that the main limiting factor is not the resolution of the images, but the conditions in which they were acquired.

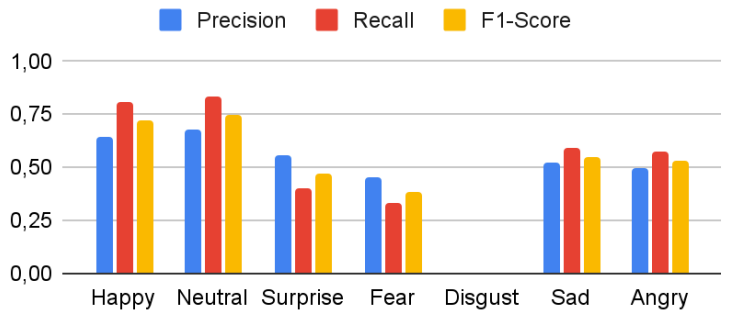
The **remaining 103 images**, not attributable to one of the specific causes mentioned above, are to be considered **model errors**.

It is hypothesized that applying **the forced detection improvement technique** to the 199 images could contribute to a more accurate classification. This approach would **reduce the number of missed images** and **improve the model's ability to correctly identify emotions**. Furthermore, a graphical representation of the error distribution, following the application of this technique, could provide further insights into the system's performance.

The following graph shows a **very similar trend** to that observed in the analysis of images where the human face was identified without the use of forced detection.

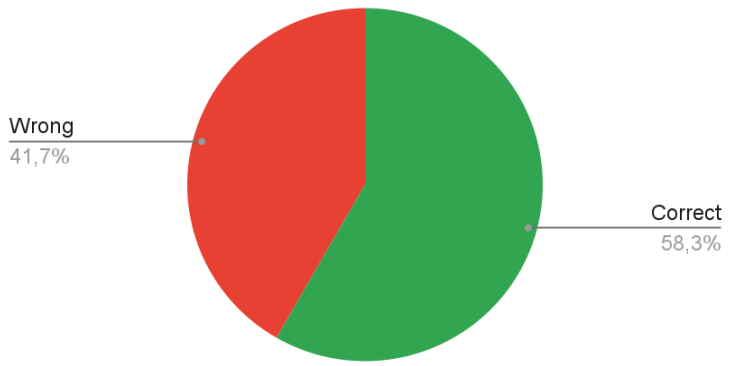


The analysis of the **precision, recall and F1-Score** metrics related to the retrieved images highlights any **drops in accuracy due to the forced detection**.

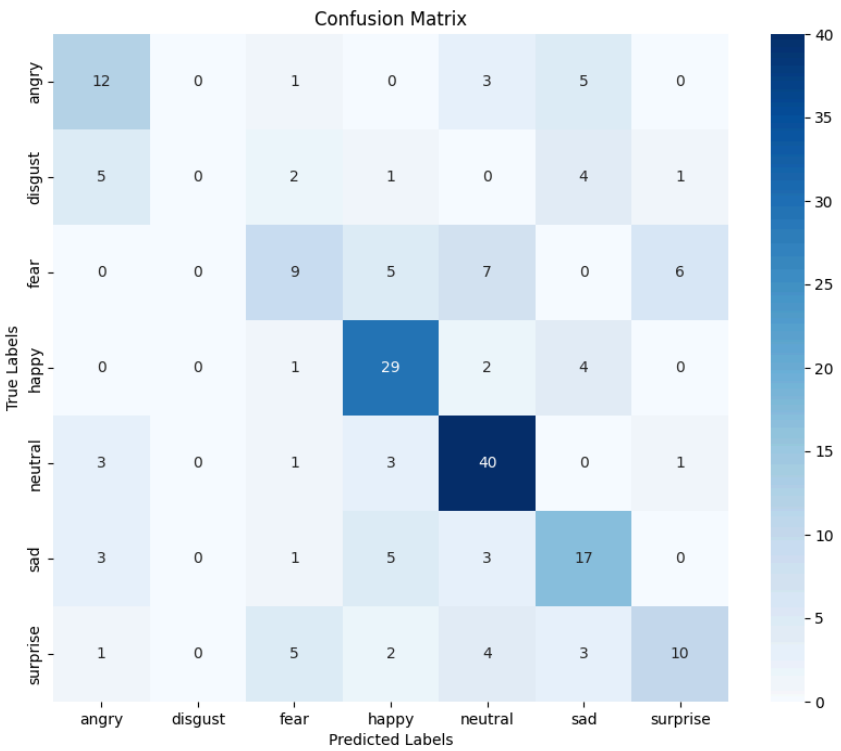


Metrics for the disgust emotion could not be calculated because all items belonging to this category were misclassified, as highlighted in the previous graph.

The percentage of misclassified images remains **stable** compared to those where the face was detected without the aid of forced detection, as can be seen in the pie chart below.



The **confusion matrix** below shows the **model's performance in classifying different emotions.**



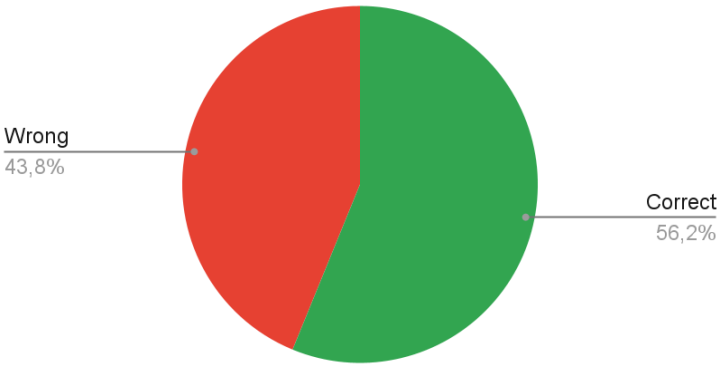
For each emotion, it is observed that:

- **Angry**: presents **12 correct classifications**, but with a **dispersion of predictions** towards other emotions, such as **sad** and **neutral**, indicating that the model may have difficulty in clearly distinguishing this emotion;
- **Disgust:** is the class with the **least number of correct predictions**, showing a **significant dispersion** between different categories, in particular **angry** and **sad**, suggesting that the model encounters particular difficulties in recognizing this emotion;
- **Fear**: is **correctly classified only in 9 cases** and presents a **significant confusion** with **neutral**, **surprise** and **happy**, highlighting possible limits of the model in distinguishing between these emotions;
- **Happy**: follows with **29 correct predictions**, but shows **classification errors** towards **sad** and, to a lesser extent, towards **neutral** and **fear**, suggesting a possible overlap in the characteristics of facial expressions;
- **Neutral**: is the class with the **best performance**, with **40 correctly classified samples**. However, **some confusions** with **happy** and **angry** are observed;
- **Sad**: shows **17 correct predictions**, but is confused with **neutral**, **happy** and **angry**, suggesting some ambiguity in the characteristics of the associated expression;
- **Surprise**: is **correctly recognized in 10 cases,** but shows a **significant dispersion** between different classes, such as **fear**, **natural** and **sad**, which could indicate difficulties in distinguishing this emotion from others with similar characteristics.

Overall, the confusion matrix shows a trend of results substantially **in line** with those emerged in the first analysis.

4.1.3 Total Enforced Analysis

In the overall analysis, considering both the images where the face was detected naturally and those subjected to forced detection, a more **significant improvement** is observed, as highlighted in the pie chart below. This result suggests that the validation set may have a **lower level of noise**.



4.2 RAF_DB Analysis

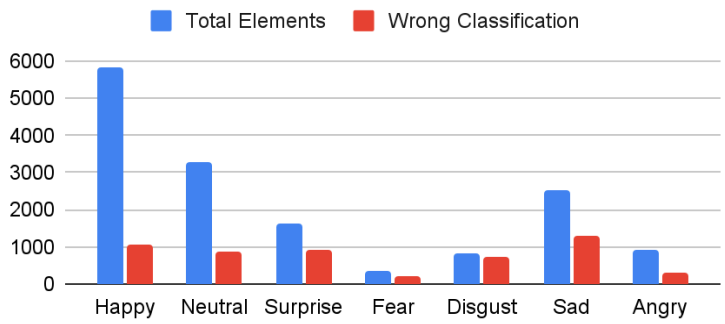
The RAF-DB dataset in the train and test folders contains a total of **15333 images**, divided into folders, one for each emotion. The emotions, as in the val_set folder of AffectNet, are divided into the following categories: **happiness (happy), neutrality (neutral), surprise (surprise), fear (fear), disgust (disgust), sadness (sad) and anger (angry).**

The distribution of the images is as follows:

- Happy: 5812 images;
- Neutral: 3271 images;
- Surprise: 1613 images;
- Fear: 370 images;
- Disgust: 847 images;
- Sad: 2518 images;
- Angry: 902 images.

4.2.1 First Analysis

The graph presented below shows the total number of items for each emotion contained in the dataset (in blue) and the amount of associated classification errors (in red).

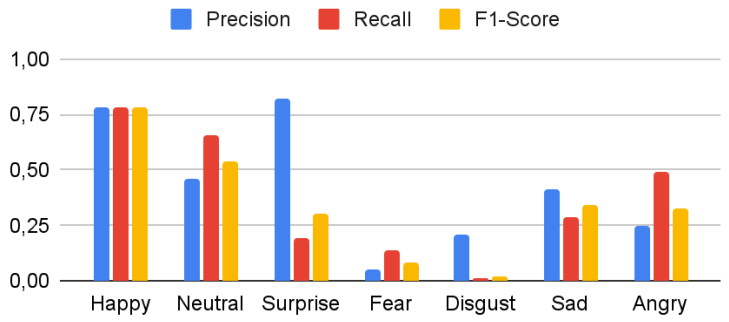


The analysis shows that:

- The most represented emotions, such as **Happy** and **Neutral**, have a **significantly high number of elements**, but also a **considerable amount of classification errors**. This suggests that, despite their large presence in the dataset, the model may have difficulty in correctly distinguishing these categories from other similar emotions;
- Some less frequent emotions, such as **Fear** and **Disgust**, show a **high error rate** compared to the total number of available elements. This indicates that the model is more difficult to recognize these categories, probably due to the **scarcity of training examples** or **greater ambiguity in the visual features** that define them;
- The emotions **Surprise** and **Sad** have an **intermediate number of elements**, but with a **relatively high percentage of errors**;
- **Angry** is among the emotions with a **lower number of errors** compared to the total number of elements, which indicates that the model is better able to identify the distinctive features of this category.

This could indicate that the neural network does **not have enough examples of these emotions in the dataset** or that the **visual features that distinguish them are not strong enough**.

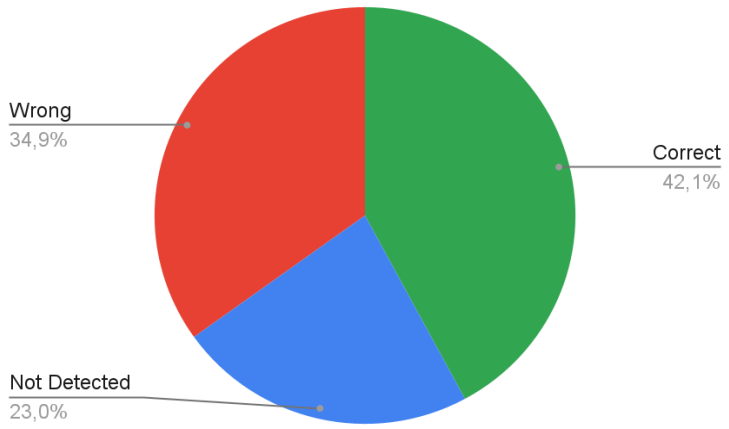
The graph below compares the **Precision**, **Recall** and **F1-Score** metrics for each emotion, giving an indication of how well the model performs in recognizing different emotional categories.



The analysis of the metrics shows that:

- **Happy and Neutral** have the **highest scores in all metrics**, indicating that the model is particularly **effective** in recognizing these emotions. This could be due to their greater representation in the training dataset or the presence of more distinctive visual features;
- **Surprise** shows a **relatively high precision**, but a lower **recall value**. This means that, although the model correctly identifies many of the instances classified as "Surprise", it tends to miss a significant number of them, incorrectly classifying them as other emotions;
- **Fear and Disgust** obtain the **lowest scores in all metrics**, highlighting a significant difficulty of the model in distinguishing these emotions. This could be due to a **poor representation in the dataset** or the **presence of less marked facial features** compared to other emotions;
- **Sad and Angry** record **intermediate values for all metrics**, indicating that the model has a moderate ability to recognize these emotions, although some ambiguity remains in distinguishing them from similar categories.

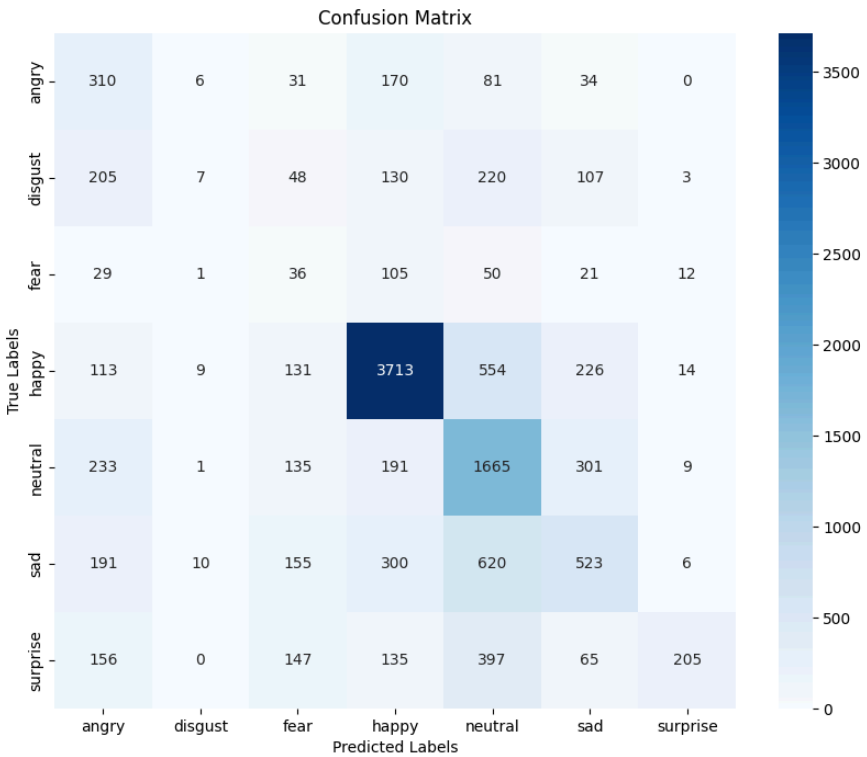
The pie chart represents the distribution of classifications performed by the model.



For each sector of the graph, it is noted that:

- **Correct Classification (42.1%):** the model correctly identifies the predominant emotion in **less than half of the samples**. Although this percentage indicates a certain degree of reliability, the relatively low value suggests the **need for further optimizations to improve the overall accuracy;**
- **Wrong Classification (34.9%):** a **considerable number of images** are incorrectly classified, highlighting the difficulty of the model in correctly distinguishing some emotions. This could be due to **similarities between facial** expressions, an **unbalanced distribution of categories in the dataset** or **intrinsic limitations of the architecture** used;
- **Not Detected (23%): almost a quarter of the analyzed images** do not have a detected face. This phenomenon could be attributed to several factors, including **partial or total occlusions of the face**, **unfavorable lighting conditions** or **insufficient image quality**.

The **confusion matrix** provides a detailed analysis of the model's performance in recognizing the different emotions.



Looking at the data, it emerges that:

- **Angry**: the model **correctly recognized 310 images**, but showed a **significant confusion** with **happy (170 cases), neutral (81 cases)** and **sad (34 cases).** This could indicate that some angry expressions are misinterpreted as more positive or neutral;
- **Disgust**: it is the class with the **least accurate predictions**, with **only 7 samples correctly classified**. It shows a **significant dispersion**, with frequent errors towards **neutral (220 cases), angry (205 cases) and happy (130 cases).** This highlights a difficulty of the model in recognizing this emotion, probably due to a **lower representation in the dataset** or a **greater variability in disgust expressions**;
- **Fear**: this emotion was **correctly identified in only 36 cases**, but presented **numerous classification errors**, in particular towards **happy (105 cases)** and **neutral (50 cases).** This indicates that the model struggles to differentiate fear expressions from other emotions with shared visual traits;
- **Happy**: This is the class with the **highest number of correct predictions**, with **3,713** samples correctly classified. However, there is **significant confusion** with **neutral (554 cases)** and, to a lesser extent, with **sad (226 cases)** and **fear (131 cases).** This suggests that some happy expressions could be interpreted by the model as more neutral or melancholic;
- **Neutral**: The model **correctly classified 1,665 samples**, but shows **significant confusion** with **sad (301 cases)** and **angry (233 cases).** This could indicate some overlap in the visual characteristics of neutral expressions compared to sad and angry ones;
- **Sad**: This emotion was **correctly identified in 523 cases**, but shows **significant dispersion**, with classification errors towards **neutral (620 cases)** and **happy (300 cases).** This suggests that the model may have difficulty clearly distinguishing sadness from other emotions with similar expressions;
- **Surprise**: the model **correctly identified 205 samples**, but showed a **strong dispersion**, with numerous incorrect classifications towards **happy (397 cases)**, **angry (156 cases)** and **fear (147 cases).** This suggests that the model has difficulty distinguishing surprise from other emotions with similar expressive characteristics.

The **confusion between similar emotions**, such as **Happy-Neutral**, **Sad-Neutral** and **Surprise-Neutral**, indicates that the model could benefit from an **improvement in the representation of the distinctive characteristics of each emotion**.

This analysis indicates that the model could benefit from a **rebalancing of the dataset**, increasing the presence of less frequent emotions, or from the **implementation of advanced techniques** to improve the classification, such as the use of more sophisticated neural networks or the use of data augmentation strategies to improve the generalization capacity of the model.

4.2.2 Enforced Analysis Not Detected

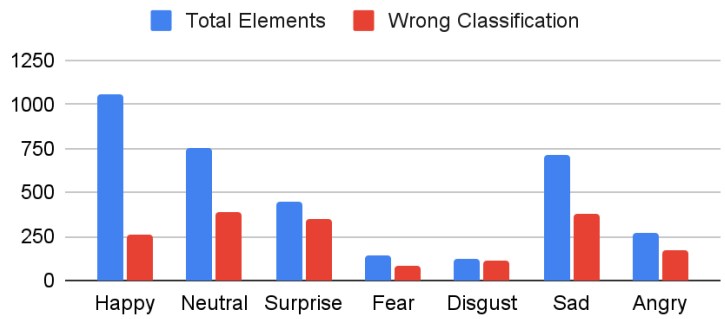
The total number of images in which the face was not detected was **1969**. For a portion of these, where possible, the cause of the failure to detect was identified, attributing it to the problems reported in the following table:

Reason for failure to detect	**Number of Images**
Lighting problems	181
Partial or total occlusions	709
Poor image quality	187
Incomplete exposure of the face	619

The analysis highlights that the **predominant cause of face failure** is represented by **partial or total occlusions, which account for approximately 36.0% of the total cases**. **Incomplete exposure of the face** follows (approximately 31.4%) and **insufficient image quality** (9.5%). Finally, **lighting problems** (approximately 9.2%), suggesting that non-optimal lighting conditions can compromise face detection, but with a lower impact than other factors.

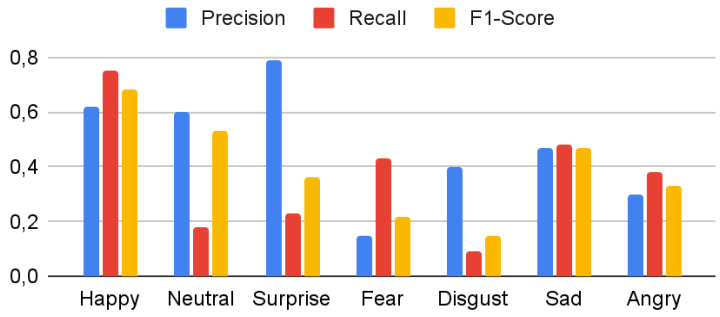
The **remaining 273 images**, not attributable to one of the specific causes mentioned above, are to be considered **model errors**.

The following graph represents the distribution of emotions in the RAF-DB dataset for images of faces initially undetected, subsequently processed using the **forced detection** technique.



The results show an **overall trend similar** to that observed in the analysis of images in which the human face was identified without resorting to forced detection, suggesting a **consistency in the model performance** regardless of the detection technique adopted.

The analysis of the **Precision**, **Recall** and **F1-Score** metrics relating to the images processed using **forced detection** highlights a **general improvement in the model performance**, in particular in the recognition of emotions that were previously more difficult to classify, although some critical issues persist, especially in the ability to correctly identify all instances of specific categories.



The emotion "**Happy**" maintains **substantially unchanged results** compared to images where the face was detected without the aid of forced detection, confirming a high **reliability** of the model in correctly classifying it on all metrics.

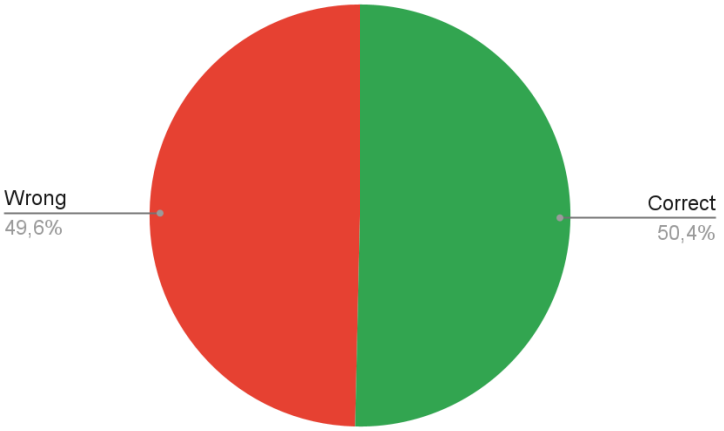
For "**Neutral**", an **increase** in **Precision** is observed, signaling a **greater accuracy** in correct classifications; however, the **decrease** in **Recall** indicates that the model identifies a **lower number of instances** than in the previous evaluation.

The emotion "**Surprise**" shows **values similar** to those recorded previously: **Precision** remains **relatively high**, while **Recall** continues to be **lower**, suggesting that, although the model accurately identifies the instances classified as "**Surprise**", a significant portion of them are not detected, resulting in classification errors.

The emotions "**Fear**", "**Disgust**" and "**Sad**", although they continue to obtain the **lowest scores** among all categories, have improved in all metrics, signaling an **increase in the reliability** of the model in recognizing them.

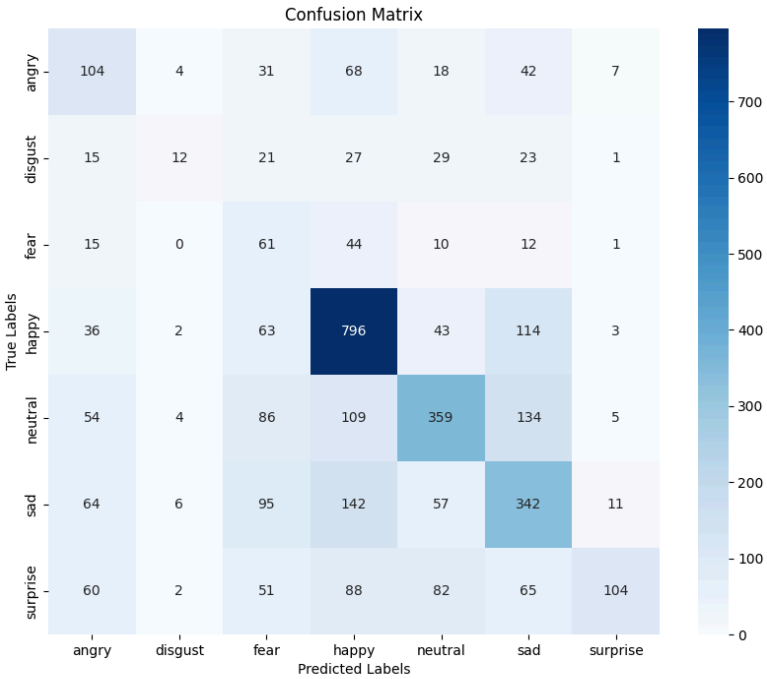
Finally, for the emotion "**Angry**", a **slight increase in Recall** is observed, which indicates a greater ability of the model to identify a **higher number of instances** of this emotion, while maintaining a **Precision** substantially **unchanged**.

The analysis of the overall distribution of the classifications performed by the model shows a **balance between the success rate and the error rate**, indicating balanced but not optimal performance, as can be seen in the pie chart below.



These results suggest the need for **further improvements**, particularly in the enfo**rcement detection** for the recognition of undetected faces, in order to increase the overall accuracy of the system.

In the **confusion matrix**, it is noted that the emotion "**Happy**" is consistently the **most recognized**, while the emotions "Fear" and "Disgust" continue to show a **significant dispersion of errors** across different categories, confirming the difficulty of the model in distinguishing them accurately.

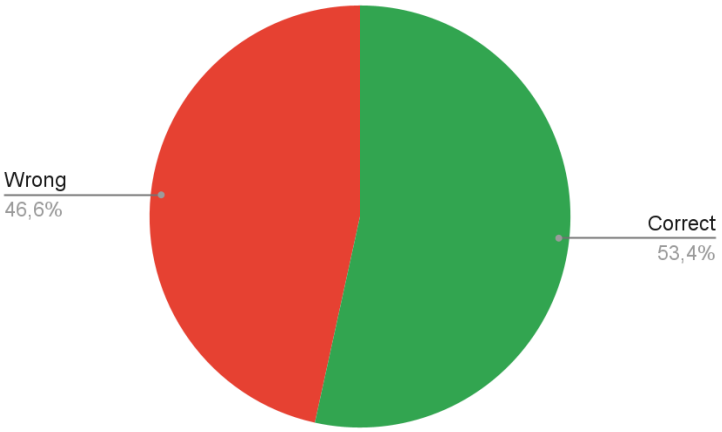


Furthermore, some differences in confusions between classes are observed compared to the first analysis:

- For the emotion "**Happy**", the **most frequent error** occurs with "**Sad**", rather than with "**Neutral**";
- The emotion "**Sad**" is **confused more often** with "**Happy**" than with "**Neutral**";
- For the emotion "**Surprise**", the **greatest confusion** is recorded with "**Happy**", as well as with "**Neutral**".

4.2.3 Total Enforced Analysis

In the overall analysis, which includes both images where the face was detected naturally and those subjected to forced detection, it emerges that the model **benefited from this technique**, especially for emotions that were previously more difficult to distinguish, as can be seen in the pie chart below.



However, **significant errors** persist, suggesting that **image quality** and the **complexity of differentiating certain expressions** continue to be critical factors affecting overall recognition accuracy..

4.3 Comparison between the two Datasets

The following considerations emerge from the analysis of the results of both datasets:

- The difficulties in recognizing emotions mainly derive from two factors: a **limited representation of certain emotional categories** within the dataset and the **similarity between some facial expressions**, which can lead to errors in distinguishing between different emotions;
- Faces are not detected due to **defects present in the images**, which can include poor lighting, non-optimal angles, partial occlusions, low resolution or partial visibility of the face, compromising the accuracy of the detection process;
- The main factor that determines the correct recognition of some emotions is their **broad representation within the training dataset**, which allows the model to learn more effectively the distinctive features of such facial expressions;
- The application of **forced detection** had a **positive impact** in both cases, although the improvement was not uniform. In particular, a **greater increase** in performance is recorded in the results related to the **val_set** folder of the AffectNet dataset. This suggests that the RAF-DB dataset has more challenges, including image quality, pose variations, and emotion distribution.

4.4 Limitations and Future Developments

The **limitations** that emerged from the analysis of the effectiveness of the model are:

- The model has a **low ability to detect complex emotions** such as **fear** and **disgust**, mainly due to the **scarcity of examples** of these emotions in the training material, which compromises its accuracy in the identification and classification phases;
- There is a **significant percentage of images where the face was not detected**, due to the presence of **defects in a part of the images present in the datasets**, such as occlusions, lighting issues, pose variations, and poor quality. These factors compromise the ability of the model to perform a correct detection, thus reducing the overall reliability of the results;
- There is a **significant amount of errors in the classification of emotions**, which indicates that the model is currently not suitable for practical applications; to be considered reliable, **several improvements** are needed in order to achieve high performance and ensure a level of accuracy sufficient for concrete use.

There are several possibilities for **future development** that can be implemented:

- **Expanding the training dataset**: including **more examples of complex emotions** (such as fear and disgust) to improve the model's ability to detect these emotions more accurately;
- **Balancing the dataset**: ensuring a **uniform distribution of images for each emotion analyzed**, in order to gain insight into the difficulties the model faces in correctly recognizing and classifying emotions;
- **Variety in image conditions**: adding **images with different poses**, **lighting**, and **quality**, in order to reduce errors related to face detection and improve the overall reliability of the model;
- **Improving preprocessing**: strengthening the robustness of the DeepFace model against occlusions and other issues common in real images, for example by implementing **preprocessing techniques** that allow it to better handle variations in quality and lighting;
- **Introduction of advanced training techniques**: use **transfer learning** and other advanced methodologies to **improve the generalization of the model**, as well as exploit data augmentation techniques to **increase the variety and quantity of training data**, thus improving the accuracy of the model;
- **Practical applications in the gambling field**: consider the adoption of the model for the **monitoring of compulsive stress in the context of gambling**, with particular attention to the **interaction with slot machines**; these developments could benefit from the concepts acquired in the **Computer Vision** and **Multimodal Interaction** courses, improving the integrated analysis of visual and acoustic signals.

Chapter 5: Conclusions

5.1 Summary of Results

The DeepFace model showed **good performance in recognizing the most distinctive and easily** identifiable emotions; however, it encountered **greater difficulties in distinguishing more complex or nuanced emotions**, as well as in non-optimal acquisition conditions, such as in the presence of occlusions, lighting variations or non-frontal angles of the face. These results suggest the need for further improvements, both in **model training** and in the **quality and diversity of the data used**. Such improvements could be achieved through an **improvement of the training dataset**, a more **advanced** network architecture or the **integration of more effective pre-processing techniques**.

5.2 Research Impact

This study highlights the need to **optimize existing models** in order to **improve the accuracy of emotion recognition**. In particular, it is essential to refine the model's ability to **distinguish similar facial expressions** and to **handle unfavorable acquisition conditions more effectively.**

5.3 Challenges Overcome

The management of undetected images and the statistical analysis of the results were addressed through **innovative strategies**, aimed at mitigating the impact of the model limitations and improving the understanding of its performance. In particular, techniques for **forced face detection** and for the **quantitative analysis of classification errors** were implemented. These approaches allowed to **precisely identify the main critical issues of the model**, highlighting the areas where improvement is needed, both in terms of detection capacity and accuracy in distinguishing emotions. These solutions also favored a more **rigorous evaluation of the overall performance**, helping to outline more targeted optimization strategies.

5.4 Closure

The project has laid the foundations for future developments in the field of **stress detection** and **emotion analysis**, opening new application perspectives in the fields of health, well-being and psychological research. The integration of increasingly accurate and robust models could favor the improvement of diagnostic tools, the monitoring of the emotional state in clinical contexts and the optimization of support systems based on artificial intelligence.

Bibliography

Deepface: https://github.com/serengil/deepface

Dataset:
AffectNet; A. Mollahosseini; B. Hasani; M. H. Mahoor, "AffectNet: A Database for Facial Expression, Valence, and Arousal Computing in the Wild," in IEEE Transactions on Affective Computing, 2017.
http://mohammadmahoor.com/affectnet/
RAF-DB; http://www.whdeng.cn/RAF/model1.html

Librerie:
https://scikit-learn.org/stable/
https://pypi.org/project/deepface/

GitHub:
https://github.com/ValerioM01
https://github.com/gabmor01

Email:
mesiti.1936543@studenti.uniroma1.it
moretti.1958932@studenti.uniroma1.it